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UbiGuard - Ubiquitous Security and Monitoring System

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Project - Deliverable 1 - Written Text

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1. Introduction.....	2
1.1. Context and Problems.....	2
1.2 Users and Roles.....	2
1.1.1 User Profile Management.....	2
1.3. Main Objectives.....	3
2. General Overview.....	3
2.1. Architecture.....	3
2.2. Ubiquitous Integration with Mobile Sensors.....	3
2.3. IoT Devices, Sensors and Actuators.....	4
2.3.1 Sensors.....	4
2.3.2 Actuators.....	4
2.4. System Functionalities from the End User's Perspective.....	5
2.4.1. Logic and Automatic Behaviors.....	5
2.4.2. Fault Tolerance and Communication Drops.....	5
2.5. Component Interaction.....	6
2.6. Demonstration Prototype.....	6
6. Bibliography.....	7

1. Introduction

Home security remains one of the most pressing concerns for residential property owners. This project consists of a complete IoT solution for residential security and automation. The system integrates a mobile application, a cloud-based server, and one or more ubiquitous sensing nodes built around ESP32 controllers (Gateways), forming an N-to-M architecture where multiple users can monitor and control multiple secured zones.

1.1. Context and Problems

Burglaries affect millions of households each year, particularly in urban and suburban areas where properties may remain unoccupied for extended periods during working hours. Even though the use of smart homes have been increasing, some of the most affordable systems lack handling of network failures, leaving homes unprotected when connectivity drops, per example: alarms that sound locally but fail to notify the homeowner remotely and systems that cannot distinguish between a resident arriving home and an unauthorized entry.

The geographic scale of this project targets a single residential unit, like a house or an apartment, where the system accommodates multiple distinct zones within a property (entrance, living area, perimeter) each equipped with its own sensing node.

1.2 Users and Roles

The mobile application will have a carefully designed graphical interface and will allow on-demand commands like arming/disarming remotely, check the temperature, check the status of the sensors and check the camera. The system is relevant because it transforms the smartphone users already carry into a context-aware security hub.

The system supports three major distinct user profiles, each with permissions that reflect household dynamics and professional installation needs.

1.1.1 User Profile Management

In our application we are going to have different types of profiles.

- **Installer Profile:** This profile has access to IoT network diagnostic tools and sensor calibration and does not grant access to camera feeds, event history, or day-to-day alarm control.
- **Owner Profile:** This is the primary administrator of the systems. It has full control, including camera feed viewing, event history, arming and disarming the system and system management.
- **Child/Guest Profile:** Both profiles offer limited functionalities, allowing them only to disarm the alarm with their personal PIN, without access to settings. The Guest Profile differentiates from the Child profile because their access

has a timer specified by the Owner, once the timer expires, the guest's PIN is automatically deactivated and the owner receives a notification.

1.3. Main Objectives

The main objectives of this project are to design and implement a complete multi-component IoT architecture connecting mobile clients, a cloud server, and sensing nodes through appropriate communication protocols; to provide security behaviours that respond to sensor data and occupant presence; to allow authorized users to monitor and control the system remotely; to enforce a role-based access model with distinct Installer, Owner, Child, and Guest profiles supporting time-bound temporary access; and tolerate fault communication drops.

2. General Overview

2.1. Architecture

UbiGuard is designed as an Edge-Cloud architecture connecting mobile clients, a cloud backend, and the physical environment. To ensure high reliability and optimize the prototype's performance, the system utilizes an **All-in-One Edge Gateway**.

Instead of a distributed multi-node network, a single ESP32 microcontroller acts simultaneously as the sensing node (connecting directly to all local sensors and actuators via GPIO) and the central gateway. This controller processes frontline logic and communicates with Firebase (our central cloud backend) via standard IoT protocols. Firebase provides real-time database synchronization and built-in user authentication, ensuring instant communication between the hardware and the Mobile App.

Local Network Fallback (Offline Mode) To address connectivity vulnerabilities, the architecture includes a local communication fallback. If the external internet connection or the Firebase server goes down, the ESP32 Gateway automatically hosts a local HTTP/WebSocket server. If the user's smartphone is connected to the same home Wi-Fi network, the Mobile App detects the cloud outage and routes commands directly to the ESP32's local IP address. This guarantees that users can still receive alerts, arm, and disarm the system using their phones even during total internet blackouts.

2.2. Ubiquitous Integration with Mobile Sensors

The application will use the smartphone's native sensors to create meaningful behaviors:

- **GPS:** Upon detecting that the Owner's device GPS has moved away from the residence, the App sends a smart notification: *"We noticed you left home. Do you want to arm the alarm?"*, if the Owner forgot to turn the alarm on.
- **Wi-Fi Detection:** Upon detecting that the user's smartphone has connected to the specific home Wi-Fi network, the App can communicate with the server to automatically disarm the system for convenience.

2.3. IoT Devices, Sensors and Actuators

The ubiquitous layer of the system is built around a gateway node based on the ESP32 microcontroller, chosen for its built-in Wi-Fi connectivity, low power consumption, and sufficient processing capability to run local automation logic during offline operation. The gateway aggregates data from all connected sensors, executes the immediate response logic locally, and communicates with the cloud server for coordination and logging.

The local system uses a diverse network, fulfilling the requirement to use at least two different types of sensors and two different types of actuators.

2.3.1 Sensors

- **Magnetic Contact Sensor (MC-38):** Detects the opening of perimeter doors/windows.
- **PIR Motion Sensor (HC-SR501):** Detects volumetric presence indoors.
- **Temperature and Humidity Sensor (DHT11):** Continuous monitoring of the indoor climate.
- **RFID Reader (RC522):** Reads physical tags for quick authentication.
- **Membrane Keypad (4x4):** Physical interface for entering local PIN codes.

2.3.2 Actuators

- **Visual Signaling Module (LEDs):** Green LED (system disarmed) and Red LED (system armed/alert).
- **OLED Display (0.96" I2C):** Main visual interface that displays dynamic text with the system's status ("System Armed", "Disarmed", "Fire Alert").
- **Audio Signaling (Active Buzzer):** High-intensity local audio alert.

- **Ubiquitous Surveillance Camera (Dedicated Smartphone):** Repurposing a mobile device placed in the environment, which acts by starting video recording/streaming after receiving a remote command from the system.

2.4. System Functionalities from the End User's Perspective

Upon arriving home, the system detects the owner's smartphone connecting to the home Wi-Fi network and can automatically disarm the alarm, deactivating the siren standby and updating the local OLED display to reflect the new state. Alternatively, a user standing at the door can disarm the system by tapping a personal RFID tag or entering a PIN on the membrane keypad, without needing to reach for the phone at all. When the owner leaves the residence, the mobile application detects the GPS location change and sends the notification, allowing the owner to arm the system with a single tap rather than triggering an automatic action that could cause accidental lockouts.

2.4.1. Logic and Automatic Behaviors

The system does not act in isolation, it presents automatic behaviors depending on external factors and the combination of multiple inputs.

- **Visual State Management:** When arming the system (via App, PIN, or RFID), the Gateway immediately changes the local interface: the Green LED turns off, the Red LED turns on, and the OLED Display updates the message to "SYSTEM ARMED".
- **Combined Intrusion Action:** If the window sensor is activated while the system is disarmed, the automatic action is simply logging the event in a file on the server. However, if the alarm is armed and there is a critical combination, window sensor followed by the PIR motion sensor triggering, the system enters a pre-alarm state, starting a 30-second countdown on the OLED Display.
- **Emergency Escalation:** If the alarm is not deactivated within that time limit, the system acts: it triggers the siren, sends the alert information to the other system components (notifying the App), and sends a remote command for the "Camera Smartphone" to wake up and start recording the room.
- **Environmental Monitoring:** The DHT11 sensor continuously checks the temperature. If it detects a sudden rise ($>60^{\circ}\text{C}$), regardless of the alarm's state, the system automatically triggers a fire alert on the OLED, sounds the siren, and notifies the Cloud.

Household members with Child profiles can disarm the system using their personal PIN on the keypad or their personal RFID tag, while Guest profiles allow temporary visitors to do the same within a time window defined by the owner.

2.4.2. Fault Tolerance and Communication Drops

A security system must be resilient. The system's behavior in the event of a network communication drop has been defined. Since the immediate logic resides in the Gateway, if the connection to the central server drops, the alarm continues to function fully autonomously locally. The OLED Display will show "Warning: Offline Mode", but the keypad code, RFID, and siren trigger rules will remain active. Event logs will be saved in the Gateway's local memory and sent in a batch to the server as soon as the connection is restored.

2.5. Component Interaction

The interaction between system components operates on a dual-layer model: a primary cloud-mediated channel and a resilient local fallback, all centered around a single, unified ESP32 controller.

- **Cloud-Mediated Interaction (Primary):** Under normal conditions, the cloud server mediates all interactions to ensure a scalable and secure architecture. The All-in-One ESP32 gateway directly reads physical inputs (PIR motion, magnetic contacts, keypad, and the RFID reader) and publishes these events and state changes to the cloud backend using MQTT. The server evaluates this data, stores events in the database, and forwards real-time updates to the connected mobile clients. Furthermore, if an intrusion is detected, the server sends a dedicated activation command to the surveillance camera smartphone, triggering it to wake up and start «streaming or recording footage.
- **Direct Local Interaction (Fallback):** To tolerate internet outages or server downtime, the system includes a direct local link. If the cloud connection drops, the ESP32 automatically hosts a local HTTP/WebSocket server. If the user's mobile app is connected to the same home Wi-Fi network, it detects the cloud outage and routes commands directly to the ESP32's local IP address. Additionally, physical authentication hardware like the **RFID reader** and keypad continue to function fully autonomously, allowing users on-site to disarm the system without external internet access.
- **Contextual Mobile Triggers:** The GPS and Wi-Fi sensors on the user's mobile device feed data to the application locally. The App evaluates this context and communicates arming suggestions (when the user leaves) or automatic disarming

commands (when the user connects to the home Wi-Fi) directly to the server, which then synchronizes the ESP32 gateway.

In summary, while the cloud server primarily handles complex automation and media routing (like the camera), the system dynamically adapts to allow direct App-to-Gateway communication over the local network when necessary.

2.6. Demonstration Prototype

For the course prototype, the following subset of components will be implemented to demonstrate the full architecture at a reduced scale:

- **1 Android mobile application** - fully functional, supporting Owner, Child, and Guest profiles with GPS-based arming notifications, Wi-Fi-based automatic disarming, push notifications, on-demand commands, and event history.
- **1 dedicated surveillance camera smartphone** - a repurposed Android device placed in the monitored environment, capable of receiving remote commands from the server to start recording.
- **Firebase Backend** - acting as a serverless infrastructure to handle the database, automation logic (via Cloud Functions), and push notifications (via Firebase Cloud Messaging). For the prototype, this provides a fully managed cloud solution, eliminating the need for local server hosting.
- **1 ESP32 gateway node** - simulates a full entry-point security station, equipped with the magnetic contact sensor (MC-38), the PIR motion sensor (HC-SR501), the RFID reader (RC522), the membrane keypad (4×4), the OLED display, the LED pair, and the active buzzer.

6. Bibliography

1. *Victimization During Household Burglary*, BJS, U.S. Department of Justice — <https://bjs.ojp.gov/content/pub/pdf/vdhib.pdf>